Stochastic Sampling is Epistemically Shallow: The Dimensionality Gap Between Temperature Variation and Model Diversity in LLMs

Izhar Ali [1]

Abstract

When a language model gives different answers on repeated runs, does that variation reveal what it does not know? Self-consistency turns the variation into a per-question uncertainty estimate via majority voting. But does the same variation reveal cross-question structure—related questions flipping together, the way a diverse ensemble does? We compare two regimes on the same questions: one model run 100 times at $\tau = 1$ versus an ensemble of 24 LLMs run once each at $\tau = 0$. A Marchenko–Pastur random-matrix test separates signal from sampling noise on both sides. Within any single model, at most one dimension rises above noise across five families and three benchmarks (MMLU, HellaSwag, GSM8K). Across the ensemble, four eigenvalues clear the noise edge, while a matched-difficulty Bernoulli null produces at most one in 500 Monte Carlo draws. Self-consistency gives accurate per-question uncertainty but no detectable cross-question structure; only a diverse ensemble surfaces what a model does not know.

1. Introduction

Self-consistency (Wang et al., 2023) has become the default low-cost uncertainty estimator for LLMs, alongside descendants that read stochastic variation as a signal of what a model knows: answer-recurrence confidence (Kadavath et al., 2022), hallucination flagging by agreement (Manakul et al., 2023), semantic entropy (Farquhar et al., 2024). Whether that signal can stand in for the more expensive alternative of deep ensembles (Lakshminarayanan et al., 2017) depends on what kind of information it carries.

First, stochastic samples can estimate a *per-question* success probability p_i by averaging independent noise. Second, they can reveal *cross-question* structure: correlated errors across related questions, the way a diverse ensemble does (Kendall & Gal, 2017; Hüllermeier & Waegeman, 2021). To substitute for an ensemble, self-consistency needs the second kind too.

Cross-model dimensionality is well characterized—a single factor captures 79% of variance on 5,000+ models (Kipnis et al., 2025), factor analysis yields 8 dimensions on 60 models (Maimon et al., 2025), low-dimensional joint embeddings (Yao et al., 2025)—but these all analyze a model×benchmark score matrix. Within-model dimensionality, under a sampling-noise null, has not been measured. We close this gap with a Marchenko–Pastur test (Marčenko & Pastur, 1967) on the run×question correctness matrix.

One dimension within, four across: this is the *dimensionality gap*.

Contributions.

1. We define within-model epistemic dimensionality as the number of independent directions of structured error in one model's stochastic samples.
2. Empirically, we find at most one above-noise dimension within each of five LLMs across three benchmarks (MMLU, HellaSwag, GSM8K chain-of-thought), but four above-noise dimensions across 24 diverse models on MMLU—unmatched by any of 500 matched-difficulty Bernoulli null draws.

2. Method

2.1. Within-Model Stochastic Probing

Let $\mathcal{M}$ be a language model and $\{q_1, \ldots, q_N\}$ be N questions. We generate K stochastic completions per question at temperature $\tau > 0$. For run k and question i, define

$$c_{ki} = \begin{cases} 1 & \text{if } \mathcal{M} \text{ answers } q_i \text{ correctly in run } k, \\ 0 & \text{otherwise.} \end{cases}$$

[1]Rowan University. Correspondence to: Izhar Ali <aliizh94@rowan.edu>.

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This gives a $K \times N$ binary matrix $\mathbf{C}$.

A question is *borderline* if its pass rate $f_i = \frac{1}{K}\sum_k c_{ki}$ lies in $(\varepsilon, 1-\varepsilon)$ with $\varepsilon = 0.05$. Let $\mathbf{C}_B$ denote $\mathbf{C}$ restricted to the N_b borderline questions; only these contribute non-trivial variance to the correlation test below.

2.2. Marchenko–Pastur (MP) Null Test

We test column independence in $\mathbf{C}_B$ using a random-matrix null. Compute the $N_b \times N_b$ correlation matrix $\mathbf{R}$ of $\mathbf{C}_B$ (each borderline question a variable, each of the K runs an observation) and its eigenvalues $\lambda_1 \geq \cdots \geq \lambda_{N_b}$. Under the null of independent Bernoulli columns, $\mathbf{R}$'s spectrum follows the Marchenko–Pastur law (Marčenko & Pastur, 1967) with upper edge $\lambda_+ = (1+\sqrt{\gamma})^2$, $\gamma := N_b/K$, and the standardized top eigenvalue $z := (\lambda_1 - \lambda_+)/\sigma_{\mathrm{TW}}$ converges to Tracy–Widom F_1 (Prop. A.2). We use z as the primary null test and report the count $|\{i : \lambda_i > \lambda_+\}|$ as a coarse summary; cross-checked against Horn's parallel analysis (Horn, 1965) (App. B.1).

Rejecting this null is direct evidence of structured internal uncertainty: temperature sampling injects independent noise per prompt, so cross-question covariation requires a coupling mechanism it does not supply. If each question flips independently, the correlation matrix is asymptotically pure noise—no eigenvalue exceeds λ_+ beyond Tracy–Widom (TW) fluctuations (Proposition A.2, App. A).

2.3. Across-Model Comparison

We apply the same MP test to a diverse ensemble of M models, one run each at $\tau = 0$, with models (rather than runs) as observations. We restrict to questions where models split (neither all-correct nor all-incorrect)—the across-model analogue of the borderline filter from §2.1. The primary statistic is the MP signal count, calibrated against an independent-Bernoulli null with per-column rates matching observed per-question pass rates (500 Monte Carlo draws; §3.3). For comparability with prior factor-analytic work (Kipnis et al., 2025; Maimon et al., 2025; Yao et al., 2025; Wen et al., 2025) we also report the Shannon effective rank $\exp(H)$; it saturates near the M–1 ceiling under any heterogeneous-rate null and is a comparability statistic, not an independence test (App. B.3).

2.4. Experimental Setup

Dataset. $N = 500$ MMLU (Hendrycks et al., 2021) questions (test split), stratified across 57 subjects.

Within-model. Five models, $K = 100$ stochastic completions per question at $\tau = 1.0$: Qwen2.5-7B-Instruct, Mistral-7B-Instruct-v0.3, SmolLM2-1.7B-Instruct, Phi-3-mini-4k-Instruct, Meta-Llama-3-8B-Instruct. Temperature robustness: $\tau \in \{0.5, 1.5\}$, $K = 30$, $N = 100$ on Qwen2.5-7B.

Across-model. $N = 500$ MMLU questions, $K = 1$ run per model at $\tau = 0$, across 24 instruction-tuned models spanning eleven families and 0.5B–22B: Qwen (5), Mistral (4), Zephyr-7B, Phi (3), SmolLM2-1.7B, OLMo-2-7B, Yi-1.5-6B, DeepSeek-7B, Granite-3.1-8B, Llama (4), Gemma-2 (2).

3. Results

3.1. Within-Model: No Structure Above Noise

Within a single model, the run×question correctness matrix is statistically indistinguishable from independent Bernoulli draws: at most one eigenvalue reaches the MP edge, and only within Tracy–Widom sampling noise. Qwen2.5-7B-Instruct (71.2% at $\tau = 1.0$; 77.4% greedy) has $121/500$ borderline questions; the remaining 379 are near-deterministic.

MP test. The top eigenvalue sits below the MP noise edge ($\lambda_1 = 4.17$ vs. $\lambda_+ = 4.41$ at $\gamma = 1.21$): zero above-noise eigenvalues (Fig. 1a). Getting a history question right on a given pass predicts nothing about a chemistry question on the same pass nor about any same-subject question (within- and between-subject borderline correlations both ≈ 0). Horn's parallel analysis agrees. The $d_{\text{within}} \leq 1$ ceiling holds across $\varepsilon \in [0.01, 0.20]$ (Fig. A1).

Replication across five families. The same pattern holds on Mistral-7B, SmolLM2-1.7B, Phi-3-mini and Meta-Llama-3-8B ($K = 100$, $\tau = 1.0$; borderline counts 121–403); SmolLM2's $\lambda_{\max} = 9.05$ touches $\lambda_+ = 9.04$ but stays within MP sampling noise (TW $z = +0.05$; Table 1). Five families, 1.7B–8B parameters, none with MP-significant signal.

3.2. Generalization: HellaSwag and GSM8K

The null is not an MMLU artifact. Three models on HellaSwag (Zellers et al., 2019) ($N = 200$, $K = 50$) and Qwen2.5-7B on GSM8K chain-of-thought (Cobbe et al., 2021) ($N = 100$, $K = 30$) also give zero signal eigenvalues (Table 1). Within- and between-category correlations on HellaSwag are near zero ($|r| \leq 0.05$): no semantic coupling even under the dataset's activity grouping.

3.3. Across-Model: Structured Multi-Dimensional Disagreement

The 24-model correctness correlation matrix has four eigenvalues above the MP edge, unmatched by any of 500 matched-difficulty independent-Bernoulli draws ($p \leq 1/500$). Model accuracies span 40.2–77.4%; on 459/500

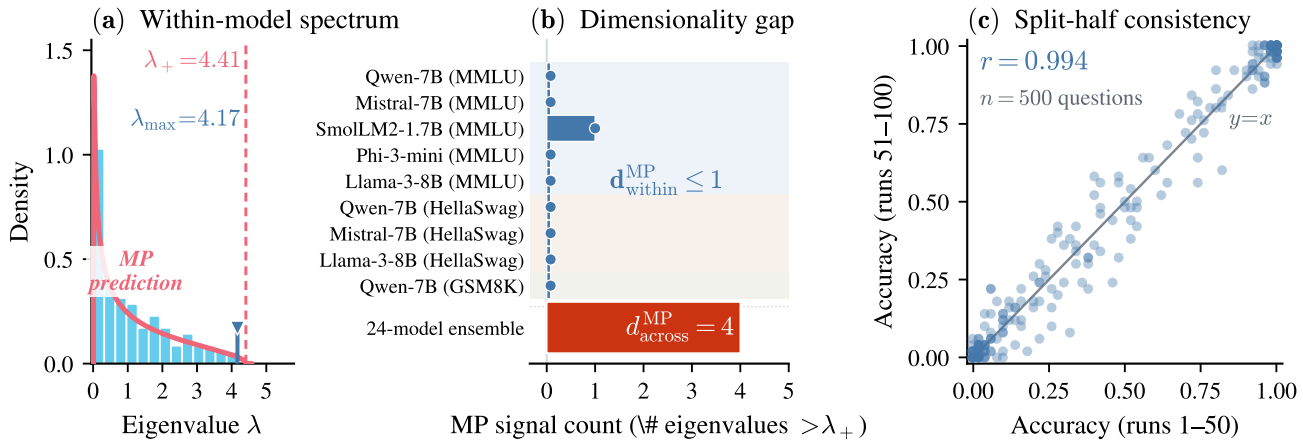


Figure 1. **The dimensionality gap. (a)** Eigenvalue spectrum of the correctness correlation matrix on borderline MMLU questions (Qwen2.5-7B, $K=100$, $N_b=121$). No eigenvalue exceeds the MP noise edge $\lambda_+ = 4.41$ (red dashed). **(b)** Same-metric comparison: MP signal count is ≤ 1 for every one of five families on three tasks but 4 for the 24-model ensemble ($\tau=0$), unmatched by any of 500 matched-difficulty independent-Bernoulli null draws. **(c)** Split-half p_i on Qwen2.5-7B ($r = 0.994$): successive samples are independent draws from a fixed per-question Bernoulli.

Table 1. Marchenko–Pastur test across three benchmarks. Margin $\lambda_+ - \lambda_{\max}$ is non-negative or within MP noise for every model (SmolLM2 ≤ 1 signal dimension). TW z: standard deviations by which $\lambda_{\max}$ exceeds the MP edge under the null (App. A); Monte Carlo mean ≈ -1.60 at $K=100$.

Model	N_b	λ_+	$\lambda_{\max}$	**margin**	**TW** z
MMLU ($K=100$, $N=500$)					
Qwen2.5-7B	121	4.41	4.17	+0.24	−2.01
Mistral-7B-v0.3	283	7.19	6.85	+0.34	−2.37
SmolLM2-1.7B	403	9.04	9.05	−0.01	+0.05
Phi-3-mini	269	6.97	6.78	+0.19	−1.36
Meta-Llama-3-8B	173	5.36	5.05	+0.31	−2.37
HellaSwag ($K=50$, $N=200$)					
Qwen2.5-7B	15	2.40	2.21	+0.18	−1.15
Mistral-7B-v0.3	58	4.31	4.04	+0.28	−1.45
Meta-Llama-3-8B	82	5.20	4.80	+0.40	−1.96
GSM8K ($K=30$, $N=100$)					
Qwen2.5-7B	65	6.11	5.72	+0.39	−1.27

Table 2. The dimensionality gap on 500 MMLU questions. Within-model columns report ranges across the five MMLU models (Table 1). $\lambda_{\max}$: top eigenvalue of the correctness correlation matrix (both sides). MP signal count is primary and separates the regimes (≤ 1 vs. 4); Shannon $\exp(H)$ saturates near its ceiling on both sides (64–87% vs. 81%).

	Within-model (5 models)	**Across-model** (ensemble)
Variation source	stochastic sampling	model identity ($\tau = 0$)
Runs / models	$K = 100$	$M = 24$
Questions	500	500
Borderline / disagree	121–403	459
Pairwise corr.	$\lvert r\rvert < .005$	.01–.63
$\lambda_{\max}$	4.17–9.05	83.0
$\lambda_{\max}/\lambda_+$	0.94–1.00	2.87
MP signal count	$\leq$ **1**	**4**
Shannon $\exp(H)$	63.5–86.0	18.7
% of ceiling	64–87% ($K-1$)	81% ($M-1$)

questions (91.8%) at least one model disagrees. Mean pairwise Pearson $r=0.35$ (range 0.01–0.63), higher on same-family pairs (Table 2; full distribution in Fig. A2).

MP count (primary). $\lambda_{1..4} = 83.0, 43.1, 36.5, 34.6$ against $\lambda_+ = 28.9$. The matched-difficulty null (§2.3; 500 draws) produced at most one signal eigenvalue on every draw (mean 0.028, max 1). Even matched for per-question difficulty, independent Bernoullis cannot produce the four above-noise directions we observe.

Shannon effective rank (continuity with prior work). The same matrix has $\exp(H) \approx 18.7$, below every null draw ($\mathbb{E}[\exp(H)] \approx 22.3$; $p \leq 1/500$). But $\exp(H)$ does not discriminate regimes: within- and across-model values both sit in a ceiling-adjacent band (Table 2; App. B.3).

3.4. Robustness and Downstream Impact

Temperature robustness. At $\tau=1.5$ ($K=30$, $N=100$, Qwen2.5-7B), $d_{\text{within}}=0$. At $\tau=0.5$ only 3 questions are borderline—below the MP test's data floor, so no conclusion is drawn at this temperature.

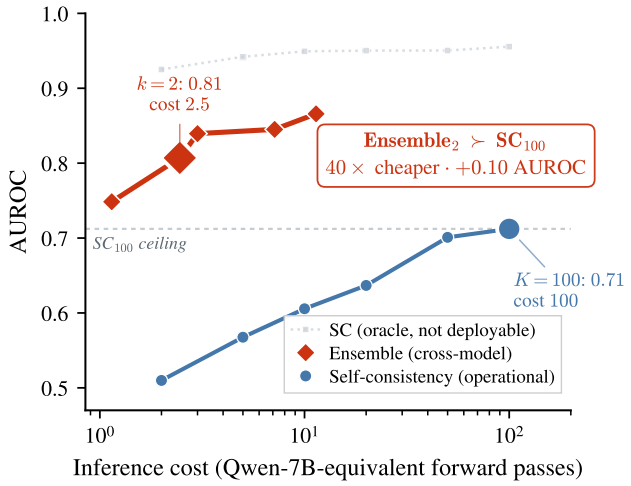


Figure 2. **Cost-AUROC Pareto frontier for selective prediction on Qwen2.5-7B ($\tau = 0$), 500 MMLU questions.** Ensemble$_2$ (AUROC 0.81) beats SC$_{100}$ (0.71) at $\sim 1/40$th the compute cost. Grey dotted: SC oracle upper bound (requires ground-truth labels at inference, not deployable). Ensemble Pareto-dominates operational SC across the entire cost range. Cost: Qwen-7B-equivalent forward passes.

Per-question estimation. Split-half p_i on Qwen2.5-7B correlates at $r = 0.994$ (Fig. 1c), calibration gap below 0.02: stochastic sampling captures each scalar p_i precisely but carries no cross-question information.

Selective prediction. Two peer models beat 100-sample self-consistency at $\sim 1/40$th the cost. The task is predicting whether Qwen2.5-7B's $\tau = 0$ answer is correct, with no ground-truth labels at inference. Operational SC$_{100}$ (fraction of the 100 samples agreeing with the modal answer) reaches AUROC 0.712.

Llama-3.1-8B + Gemma-2-9B, each run once at $\tau = 0$, reach 0.807 at $\sim 1/40$th the cost (Qwen-7B-equivalent forward passes; App. C). A single external model (Llama-3.1-8B) edges past SC$_{100}$ at $\sim 1/88$th the cost (AUROC 0.749 vs. 0.712), though this single-peer gap is marginal at $N = 500$ (DeLong $p \approx 0.2$); the two-peer gap is highly significant ($p < 0.005$).

4. Discussion

Why temperature variation is shallow. Temperature is a uniform dilation of the logits: it does not selectively modulate any knowledge domain. Consistent with this, within- and between-subject correlations are both ≈ 0 (§3.1), and TW z-scores across five models and three benchmarks are individually consistent with the null (all $z \leq +0.05$; Prop. A.2, Table 1).

Locally committed, globally incoherent. When Qwen2.5-7B errs on a borderline question, it picks the same wrong answer 87.4% of the time across samples—yet these wrong-answer preferences do not associate across questions (Appendix D). The model has strong local commitments; it just doesn't chain them into global structure.

Exception: SmolLM2. SmolLM2 reaches TW $z \approx +1.77$ at $\varepsilon = 0.15$ (∼99th percentile), consistent with slight residual cross-question coupling that Assumption A.1 idealizes away; a candidate mechanism is chat-template randomness.

Implications for epistemic intelligence. Selective-prediction research should budget for diverse models, not deeper sampling. Stacking more samples from a single model cannot recover a dimensionality that is not there.

5. Limitations and Future Work

- *Binary correctness.* We test for structure in correctness; richer signals (log-probabilities, semantic clusters) could carry structure that correctness misses.
- *MP power.* The test operates at $\gamma \in [0.3, 4.0]$; $K \gg N_b$ would sharpen detection of structure near the MP edge.
- *Across-model ceiling.* d_{across} has not asymptoted by $M = 24$, so "four" is a lower bound on the true rank.
- *Post-training.* Instruction-tuned models may differ from pretrained-only bases.

Future work.

- *Scope.* Open-generation benchmarks (TruthfulQA, HaluEval) and a 70B-tier within-model test.
- *Richer probes.* Log-probability or semantic-cluster within-model null; prompt perturbation as a second stochastic axis.
- *Mechanism.* Decompose the same-family SC$_{100}$ win (§C.3) into disagreement vs. calibration; compare against a multi-seed Pythia ensemble (Lakshminarayanan et al., 2017).
- *Matched sweep.* Temperature sweep at matched K to separate sample-count from temperature effects.

6. Conclusion

Per-question uncertainty lives within a model ($r = 0.994$); multi-dimensional, cross-question uncertainty lives between them. Temperature sampling cannot couple errors across questions; model diversity can—and two peer models outperform 100-sample operational self-consistency at $1/40$th the cost (§3.4). To go beyond a single scalar per question, ask a different model.

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A. Null Prediction and Test Statistic

This appendix formalizes the asymptotic null behavior of the MP test from §2.2.

Assumption A.1 (Per-question independence). Assume $c_{ki} \sim \mathrm{Ber}(p_i)$ are mutually independent across questions i within a run and across runs k. Operationally: independent forward passes with fresh sampling RNG per question and no shared KV cache, as in the standard eval pipeline used here.

Proposition A.2 (Null spectrum and test statistic)**.** *Under Assumption A.1 with $p_i \in [\varepsilon, 1-\varepsilon]$ for all borderline questions: (i) the population Pearson correlation matrix of $\mathbf{C}_B$'s columns is $\mathbf{I}_{N_b}$; (ii) as $K \to \infty$ with γ fixed, $\mathbf{R}$'s empirical spectrum converges to Marchenko–Pastur on $[\lambda_-, \lambda_+]$ and the standardized top eigenvalue $z = (\lambda_1 - \lambda_+)/\sigma_{\mathrm{TW}}$ converges in distribution to Tracy–Widom F_1, with edge scale $\sigma_{\mathrm{TW}} = (1+\sqrt{\gamma})(1+1/\sqrt{\gamma})^{1/3}K^{-2/3}$. z is therefore the appropriate null test; the unbuffered count $|\{i : \lambda_i > \lambda_+\}|$ has a non-vanishing $O(1)$ limit (since $\mathbb{P}(\mathrm{TW}_1 > 0)$ is bounded away from zero) and is reported as a coarse summary only.*

Sketch. Independence gives $\mathrm{Cov}(c_{ki}, c_{kj}) = 0$ for $i \neq j$ and $p_i(1-p_i) > 0$ on borderline questions, so the population Pearson correlation is $\mathbf{I}_{N_b}$. By Hoeffding with a union bound over the N candidate questions, $\sup_i |f_i - p_i| \to 0$ a.s. as $K \to \infty$, so the data-driven borderline set $\{i : f_i \in (\varepsilon, 1-\varepsilon)\}$ coincides with its population counterpart with probability $\to 1$. For (ii), center by p_i (harmless because sample Pearson correlation is shift-invariant) to obtain mean-zero bounded entries with column-heterogeneous variance $p_i(1-p_i)$; this fits Pillai & Yin (2012) (MP bulk; Theorem 1.1 allows column-dependent variances and sub-exponential entries, extending Marčenko & Pastur, 1967) and Bao et al. (2012) (Tracy–Widom edge for sample correlation matrices). □

B. Supplementary Analyses

B.1. Parallel Analysis

As a complementary test to MP, we apply Horn's parallel analysis (Horn, 1965). We generate 100 surrogate $K \times N_b$ binary matrices with independent columns matched to $\mathbf{C}_B$ in shape and per-question pass rate, and compute each surrogate's eigenvalues. For each rank j, we compare $\lambda_j(\mathbf{R})$ to the 95th percentile of λ_j across surrogates. The reported PA count is the largest r such that λ_j exceeds its surrogate threshold for every $j \leq r$ (the classical contiguous rule). PA agrees with MP on every model and benchmark reported in the main text.

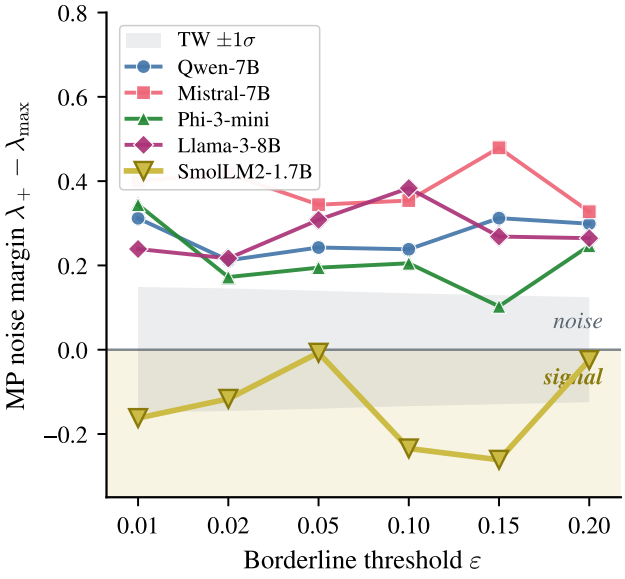


Figure A1. **Robustness to borderline threshold.** MP noise margin $\lambda_+ - \lambda_{\max}$ plotted against borderline threshold $\varepsilon \in \{0.01, 0.02, 0.05, 0.10, 0.15, 0.20\}$ for the five MMLU models ($K = 100$). Positive margin: no signal above the MP noise edge ($d_{\text{within}} = 0$); negative margin: at least one above-noise eigenvalue. Grey band: Tracy–Widom $\pm 1\sigma$ (averaged across models per ε), the typical MP-null fluctuation scale of the margin. Four of five models (Qwen, Mistral, Phi-3, Llama-3) stay strictly in the noise regime across the full ε-range. SmolLM2 sits in the signal band throughout (margin -0.26 to -0.01) but remains within $\approx 1\,\sigma_{\mathrm{TW}}$ of zero at $\varepsilon \in \{0.05, 0.20\}$: at most one above-noise dimension. The claim $d_{\text{within}} \leq 1$ is not sensitive to the choice of ε.

B.2. Robustness to borderline threshold ε

The main text uses $\varepsilon = 0.05$ to define borderline questions. Figure A1 sweeps ε across $\{0.01, 0.02, 0.05, 0.10, 0.15, 0.20\}$ and confirms the $d_{\text{within}} \leq 1$ conclusion is not an artifact of this choice.

B.3. Shannon Effective Rank: Definition and Extended Analysis

Definition. The Shannon effective rank of the correctness correlation matrix is $d_{\text{across}} = \exp(H)$, where $H = -\sum_j \hat{\lambda}_j \log \hat{\lambda}_j$ and $\hat{\lambda}_j = \lambda_j / \sum_\ell \lambda_\ell$ are the normalized eigenvalues.

Ceiling saturation. Under any heterogeneous-rate Bernoulli null, $\exp(H)$ saturates near the $M-1$ ceiling (the -1 comes from column-demeaning: centered columns lie in the $(M-1)$-dimensional hyperplane orthogonal to $\mathbf{1}_M$). Empirically, the matched-difficulty null for $M = 24$ gives $\mathbb{E}[\exp(H)] \approx 22.3$ (min 22.1). Within-model values reach 64–87% of the $K-1$ ceiling (Table 2), indistinguishable

from the across-model 81%. $\exp(H)$ therefore does not discriminate the regimes that MP counting separates cleanly.

C. Selective Prediction: Full Results

C.1. Method

Target. For each of $N = 500$ MMLU questions we predict whether Qwen2.5-7B-Instruct answers correctly at $\tau = 0$ (greedy decoding). The base accuracy is 0.774 (387/500 correct), so the trivial constant predictor achieves AUPR ≈ 0.774 and AUROC 0.5.

Operational SC$_K$ (deployable). For each $K \in \{2, 5, 10, 20, 50, 100\}$ we draw K stochastic samples from Qwen-7B ($\tau = 1.0$, the same passes used in §3.1) and define the confidence as the fraction of samples agreeing with the modal answer: $\hat{p}_i = \max_a \frac{1}{K} \sum_{k=1}^{K} \mathbf{1}[a_{ki} = a]$. This is the standard self-consistency confidence; it does not require ground-truth labels and is therefore deployable.

Oracle SC$_K$ (not deployable, reported for completeness). The oracle confidence is the fraction of the K samples that match the gold answer: $\hat{p}_i^{\text{oracle}} = \frac{1}{K} \sum_k c_{ki}$. This is an upper bound on any sample-based confidence and requires test-time label access; we report it at $K = 2$ and $K = 100$ only (Table A1), to bracket what self-consistency *could* achieve in principle.

Ensemble$_k$ (deployable). For k other models $\{\mathcal{M}_1, \ldots, \mathcal{M}_k\}$ run once at $\tau = 0$, the confidence is the fraction whose answer independently matches Qwen-7B's $\tau = 0$ answer: $\hat{p}_i^{\text{ens}} = \frac{1}{k} \sum_{j=1}^{k} \mathbf{1}[a_j(q_i) = a_{\text{Qwen}}(q_i)]$. This needs no ground truth—only Qwen-7B's own greedy answer and the k external models' greedy answers—so it is deployable today.

Cost. We report inference cost in *Qwen-7B-equivalent forward passes*: a single greedy pass of a B-parameter model costs $B/7$ units. SC$_K$ on Qwen-7B costs K units; Ensemble$_k$ costs $\sum_{j=1}^{k} B_j/7$. This isolates the parameter-FLOPs comparison from sequence-length and serving-stack effects that vary across deployments.

Selection rules. *Family-diverse*: descend by $\tau = 0$ MMLU accuracy, accepting one model per distinct family (the 11 families enumerated in §2.4). *Top-k by accuracy*: descend by $\tau = 0$ MMLU accuracy without family deduplication. We use family-diverse selection in the main figure to avoid trivial wins from stacking near-identical models; §C.3 shows the family-diversity bonus is small but real.

Table A1. Full selective-prediction results: AUROC and AUPR for predicting Qwen2.5-7B's $\tau = 0$ correctness on 500 MMLU questions. Cost is in Qwen-7B-equivalent forward passes (parameters/7 B). The oracle SC$_K$ rows are upper bounds that require label access at inference time and are shown for context only. Operational SC$_K$ uses agreement-with-modal-answer confidence; Ensemble$_k$ uses agreement with Qwen-7B's $\tau = 0$ answer over k other models. Ensemble rows are cumulative: rising k adds the next-highest-$\tau = 0$-MMLU-accuracy model under each selection rule (family-diverse: next distinct family; top-k: next model overall). The $k = 1$ family-diverse member is Llama-3.1-8B (the top non-Qwen model).

Predictor	K/k	AUROC	AUPR	Cost
Self-consistency, operational				
SC$_K$	2	0.510	0.778	2.00
SC$_K$	5	0.568	0.801	5.00
SC$_K$	10	0.606	0.818	10.00
SC$_K$	20	0.637	0.834	20.00
SC$_K$	50	0.701	0.870	50.00
SC$_K$	100	0.712	0.880	100.00
Self-consistency, oracle upper bound				
SC$_K$	2	0.925	0.964	2.00
SC$_K$	100	0.956	0.986	100.00
Ensemble$_k$, family-diverse				
Ens.	1	0.749	0.874	1.14
Ens.	2	0.807	0.905	2.46
Ens.	3	0.839	0.924	3.00
Ens.	5	0.845	0.930	7.14
Ens.	10	0.866	0.948	11.39
Ensemble$_k$, top-k by accuracy				
Ens.	5	0.880	0.946	4.43
Ens.	10	0.888	0.957	10.30

Metrics. AUROC (rank-correctness of confidence) and AUPR (area under precision-recall, sensitive to the positive base rate of 0.774). We report both but emphasize AUROC because it is invariant to the positive prevalence and thus directly comparable across setups with different base accuracies.

C.2. Full AUROC/AUPR Table

Table A1 reports every predictor at every budget we swept, with cost in Qwen-7B-equivalent forward passes. Two patterns stand out: (i) operational SC$_K$ improves slowly with K (+0.20 AUROC over a 50× cost range) and plateaus well below the cheapest ensemble—the marginal gain from $K = 50$ to $K = 100$ is only +0.011 AUROC; (ii) Ensemble$_k$ Pareto-dominates operational SC$_K$ at every cost. Top-k selection slightly beats family-diverse selection at matched k (because it concentrates parameter mass on the best models); family-diverse selection costs fewer parameters per k and remains above the SC frontier.

C.3. Family-Diversity Ablation

Does ensemble gain require diverse architectures, or is any collection of three independent runs enough? We compare two $k=3$ ensembles selected from the 24-model pool:

- *Same-family* (Qwen2.5-0.5B, 1.5B, 3B): AUROC 0.811, AUPR 0.909, cost 0.71 Qwen-7B units.
- *Different-family* (Llama-3.1-8B, Gemma-2-9B, Phi-3-mini): AUROC 0.839, AUPR 0.924, cost 3.00 Qwen-7B units.

Family diversity contributes $+0.028$ AUROC—real but modest, on top of a much larger gap between either ensemble and operational SC_{100} (0.712). Even three same-family small Qwens, at $1/140$ of SC_{100}'s cost, beat $K=100$ operational SC by $+0.099$ AUROC. This is the strongest available reading of the dimensionality gap as a deployment claim: the cross-model signal is so much richer than the within-model signal that even within-*family*-but-across-*scale* disagreement clears the SC_{100} bar by a wide margin.

D. Strong Local Preferences, Zero Global Coherence

A question is *borderline* here as in §2.1: its pass rate over the $K = 100$ samples lies in $(0.05, 0.95)$. When Qwen2.5-7B errs on such a question, it picks the same wrong choice 87.4% of the time (mean across borderline questions, $K = 100$). Yet these preferences are independent across questions: Cramér's V and mutual-information permutation tests find no pairwise associations above the 99th percentile of the null, across all five models (1,242 borderline questions, 200 permutations per model). Locally committed, globally incoherent.

E. Pairwise Model Correlations

Fig. A2 expands the across-model pairwise-correlation summary in §3.3 (mean $r=0.35$, range 0.01–0.63) to the full distribution over all $\binom{24}{2}=276$ model pairs, split by same-family vs. cross-family.

F. Extended Related Work

Across-model dimensionality. Cross-model competence dimensionality has been studied via 1D item response theory (IRT), with a single factor capturing 79% of variance on 5,000+ models (Kipnis et al., 2025); 8-factor analysis on 60 models (Maimon et al., 2025); low-dimensional joint embeddings (Yao et al., 2025); and 18 cognitive rubrics (Zhou et al., 2025). We complement these with the first structural measurement of *within-model* dimensionality across stochastic samples.

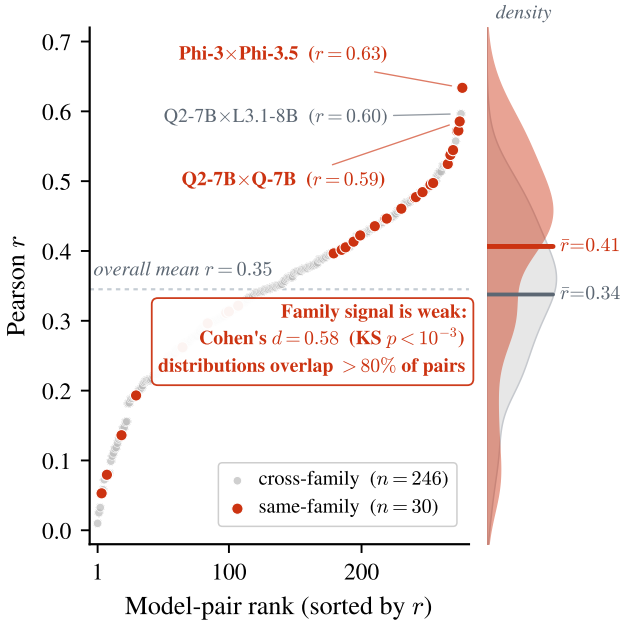


Figure A2. **Pairwise model correlations.** All $\binom{24}{2}=276$ model-pairs on MMLU ($\tau=0$), sorted by Pearson r and split by same-family (red, $n = 30$) vs. cross-family (grey, $n = 246$). Side panel: kernel density with mean bars $\bar{r}=0.41$ (same) and 0.34 (cross). Same-family pairs trend higher (Cohen's $d=0.58$, KS $p<10^{-3}$), but the same-family and cross-family densities overlap substantially. The top-3 pairs by r are annotated with their model names.

Self-consistency structure. Hamidieh et al. (2025) showed empirically that cross-model disagreement captures uncertainty self-consistency misses; we find the analogous gap structurally in the correctness covariance. Kim et al. (2025) studied pairwise error correlations across 350+ models; we measure the full factor structure. Wen et al. (2025) applied factor analysis to within-question answer-choice variation, not the cross-question covariance we analyze. Cecere et al. (2025) proposed Monte Carlo temperature; none tested here produces above-noise structure. MP is standard in portfolio theory; its application to within-model LLM uncertainty is, to our knowledge, novel in this context.